

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus:

Midrand

Faculty:

IT

Module Code:

ITDSA

Group:

2

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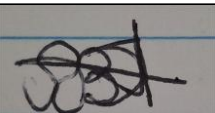
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Indicate	Yes	No
Plagiarism report attached		

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--	---------------------------

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Marks Awarded: %

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Question 1

1.1

-- PROVINCES TABLE

```
CREATE TABLE PROVINCES (  
    provinceID NUMBER PRIMARY KEY,  
    provinceName VARCHAR2(100) NOT NULL  
);
```

-- MUNICIPALITIES TABLE

```
CREATE TABLE MUNICIPALITIES (  
    muniID NUMBER PRIMARY KEY,  
    muniCode VARCHAR2(20) NOT NULL,  
    avgPopulation NUMBER,  
    provinceID NUMBER NOT NULL,  
    CONSTRAINT fk_province FOREIGN KEY (provinceID)  
        REFERENCES PROVINCES(provinceID)  
        ON DELETE CASCADE  
);
```

-- FACILITIES TABLE

```
CREATE TABLE FACILITIES (  
    facilityID NUMBER PRIMARY KEY,  
    name VARCHAR2(100) NOT NULL,  
    capacity NUMBER,  
    address VARCHAR2(150),  
    muniID NUMBER NOT NULL,
```

```

CONSTRAINT fk_muni FOREIGN KEY (muniID)
    REFERENCES MUNICIPALITIES(muniID)
    ON DELETE CASCADE
);

-- ROOMS TABLE
CREATE TABLE ROOMS (
    roomID NUMBER PRIMARY KEY,
    roomName VARCHAR2(100) NOT NULL,
    facilityID NUMBER NOT NULL,
    CONSTRAINT fk_facility_room FOREIGN KEY (facilityID)
        REFERENCES FACILITIES(facilityID)
        ON DELETE CASCADE
);

-- ACTIVITIES TABLE
CREATE TABLE ACTIVITIES (
    activityID NUMBER PRIMARY KEY,
    activityName VARCHAR2(100) NOT NULL
);

-- USES TABLE (Intersection table)
CREATE TABLE USES (
    facilityID NUMBER NOT NULL,
    activityID NUMBER NOT NULL,
    activityDate DATE NOT NULL,
    CONSTRAINT pk_uses PRIMARY KEY (facilityID, activityID, activityDate),
    CONSTRAINT fk_uses_facility FOREIGN KEY (facilityID)

```

```

REFERENCES FACILITIES(facilityID)

ON DELETE CASCADE,

CONSTRAINT fk_uses_activity FOREIGN KEY (activityID)

REFERENCES ACTIVITIES(activityID)

ON DELETE CASCADE

);

```

```

SQL> CREATE TABLE PROVINCES (
2     provinceID NUMBER PRIMARY KEY,
3     provinceName VARCHAR2(100) NOT NULL
4 );

```

Table created.

```

SQL> CREATE TABLE MUNICIPALITIES (
2     muniID NUMBER PRIMARY KEY,
3     muniCode VARCHAR2(20) NOT NULL,
4     avgPopulation NUMBER,
5     provinceID NUMBER NOT NULL,
6     CONSTRAINT fk_province FOREIGN KEY (provinceID)
7         REFERENCES PROVINCES(provinceID)
8         ON DELETE CASCADE
9 );

```

Table created.

```

SQL> CREATE TABLE FACILITIES (
2     facilityID NUMBER PRIMARY KEY,
3     name VARCHAR2(100) NOT NULL,
4     capacity NUMBER,
5     address VARCHAR2(150),
6     muniID NUMBER NOT NULL,
7     CONSTRAINT fk_muni FOREIGN KEY (muniID)
8         REFERENCES MUNICIPALITIES(muniID)
9         ON DELETE CASCADE
10 );

```

Table created.

```

SQL> CREATE TABLE ROOMS (
2     roomID NUMBER PRIMARY KEY,
3     roomName VARCHAR2(100) NOT NULL,
4     facilityID NUMBER NOT NULL,
5     CONSTRAINT fk_facility_room FOREIGN KEY (facilityID)
6         REFERENCES FACILITIES(facilityID)
7         ON DELETE CASCADE
8 );

```

Table created.

```

SQL> CREATE TABLE ACTIVITIES (
2     activityID NUMBER PRIMARY KEY,
3     activityName VARCHAR2(100) NOT NULL
4 );

```

Table created.

```

SQL> CREATE TABLE USES (
2     facilityID NUMBER NOT NULL,
3     activityID NUMBER NOT NULL,
4     activityDate DATE NOT NULL,
5     CONSTRAINT pk_uses PRIMARY KEY (facilityID, activityID, activityDate),
6     CONSTRAINT fk_uses_facility FOREIGN KEY (facilityID)
7         REFERENCES FACILITIES(facilityID)
8         ON DELETE CASCADE,
9     CONSTRAINT fk_uses_activity FOREIGN KEY (activityID)
10        REFERENCES ACTIVITIES(activityID)
11        ON DELETE CASCADE
12 );

```

Table created.

1.2

-- Sequence for PROVINCES table

```
CREATE SEQUENCE province_seq
```

```
START WITH 1
```

```
INCREMENT BY 1
```

```
NOCACHE
```

```
NOCYCLE;
```

-- Sequence for MUNICIPALITIES table

```
CREATE SEQUENCE muni_seq
```

```
START WITH 1
```

```
INCREMENT BY 1
```

```
NOCACHE
```

```
NOCYCLE;
```

-- Sequence for FACILITIES table

```
CREATE SEQUENCE facility_seq
```

```
START WITH 1
```

```
INCREMENT BY 1
```

```
NOCACHE
```

```
NOCYCLE;
```

-- Sequence for ROOMS table

```
CREATE SEQUENCE room_seq
```

```
START WITH 1
```

```
INCREMENT BY 1
```

```
NOCACHE
```

```
NOCYCLE;
```

-- Sequence for ACTIVITIES table

CREATE SEQUENCE activity_seq

START WITH 1

INCREMENT BY 1

NOCACHE

NOCYCLE;

```
SQL> CREATE SEQUENCE province_seq
2 START WITH 1
3 INCREMENT BY 1
4 NOCACHE
5 NOCYCLE;
```

Sequence created.

```
SQL> CREATE SEQUENCE muni_seq
2 START WITH 1
3 INCREMENT BY 1
4 NOCACHE
5 NOCYCLE;
```

Sequence created.

```
SQL> CREATE SEQUENCE facility_seq
2 START WITH 1
3 INCREMENT BY 1
4 NOCACHE
5 NOCYCLE;
```

Sequence created.

```
SQL> CREATE SEQUENCE room_seq
2 START WITH 1
3 INCREMENT BY 1
4 NOCACHE
5 NOCYCLE;
```

Sequence created.

```
SQL> CREATE SEQUENCE activity_seq
2 START WITH 1
3 INCREMENT BY 1
4 NOCACHE
5 NOCYCLE;
```

Sequence created.

1.3

```
INSERT INTO PROVINCES (provinceID, provinceName)
```

```
VALUES (province_seq.NEXTVAL, 'Gauteng');
```

```
INSERT INTO PROVINCES (provinceID, provinceName)
```

```
VALUES (province_seq.NEXTVAL, 'Limpopo');
```

```
INSERT INTO MUNICIPALITIES (muniID, muniCode, avgPopulation, provinceID)
```

```
VALUES (muni_seq.NEXTVAL, 'GT001', 4500000, 1);
```

```
INSERT INTO MUNICIPALITIES (muniID, muniCode, avgPopulation, provinceID)
```

```
VALUES (muni_seq.NEXTVAL, 'LP002', 3800000, 2);
```

```
INSERT INTO FACILITIES (facilityID, name, capacity, address, muniID)
```

```
VALUES (facility_seq.NEXTVAL, 'Soweto Theatre', 800, 'Soweto, Gauteng', 1);
```

```
INSERT INTO FACILITIES (facilityID, name, capacity, address, muniID)
```

```
VALUES (facility_seq.NEXTVAL, 'Polokwane Art Hall', 1200, 'Polokwane CBD', 2);
```

```
INSERT INTO ROOMS (roomID, roomName, facilityID)
```

```
VALUES (room_seq.NEXTVAL, 'Main Hall', 1);
```

```
INSERT INTO ROOMS (roomID, roomName, facilityID)
```

```
VALUES (room_seq.NEXTVAL, 'Gallery Room', 2);
```

```
INSERT INTO ACTIVITIES (activityID, activityName)
VALUES (activity_seq.NEXTVAL, 'IsZulu Dance Competition');
```

```
INSERT INTO ACTIVITIES (activityID, activityName)
VALUES (activity_seq.NEXTVAL, 'Live Jazz Concert');
```

```
INSERT INTO USES (facilityID, activityID, activityDate)
VALUES (1, 1, TO_DATE('2024-02-15', 'YYYY-MM-DD'));
```

```
INSERT INTO USES (facilityID, activityID, activityDate)
VALUES (2, 2, TO_DATE('2024-03-20', 'YYYY-MM-DD'));
```

```
SQL> INSERT INTO PROVINCES (provinceID, provinceName)
2 VALUES (province_seq.NEXTVAL, 'Gauteng');
1 row created.

SQL> INSERT INTO PROVINCES (provinceID, provinceName)
2 VALUES (province_seq.NEXTVAL, 'Limpopo');
1 row created.

SQL> INSERT INTO MUNICIPALITIES (muniID, muniCode, avgPopulation, provinceID)
2 VALUES (muni_seq.NEXTVAL, 'GT001', 4500000, 1);
1 row created.

SQL> INSERT INTO MUNICIPALITIES (muniID, muniCode, avgPopulation, provinceID)
2 VALUES (muni_seq.NEXTVAL, 'LP002', 3800000, 2);
1 row created.

SQL> INSERT INTO FACILITIES (facilityID, name, capacity, address, muniID)
2 VALUES (facility_seq.NEXTVAL, 'Soweto Theatre', 800, 'Soweto, Gauteng', 1);
1 row created.

SQL> INSERT INTO FACILITIES (facilityID, name, capacity, address, muniID)
2 VALUES (facility_seq.NEXTVAL, 'Polokwane Art Hall', 1200, 'Polokwane CBD', 2);
1 row created.

SQL> INSERT INTO ROOMS (roomID, roomName, facilityID)
2 VALUES (room_seq.NEXTVAL, 'Main Hall', 1);
1 row created.

SQL> INSERT INTO ROOMS (roomID, roomName, facilityID)
2 VALUES (room_seq.NEXTVAL, 'Gallery Room', 2);
1 row created.

SQL> INSERT INTO ACTIVITIES (activityID, activityName)
2 VALUES (activity_seq.NEXTVAL, 'IsZulu Dance Competition');
1 row created.

SQL> INSERT INTO ACTIVITIES (activityID, activityName)
2 VALUES (activity_seq.NEXTVAL, 'Live Jazz Concert');
1 row created.

SQL> INSERT INTO USES (facilityID, activityID, activityDate)
2 VALUES (1, 1, TO_DATE('2024-02-15', 'YYYY-MM-DD'));
1 row created.

SQL> INSERT INTO USES (facilityID, activityID, activityDate)
2 VALUES (2, 2, TO_DATE('2024-03-20', 'YYYY-MM-DD'));
1 row created.
```

1.4

```
SELECT COUNT(DISTINCT m.muniID) AS num_municipalities_without_music
FROM MUNICIPALITIES m
WHERE NOT EXISTS (
    SELECT 1
    FROM FACILITIES f
    JOIN USES u ON f.facilityID = u.facilityID
    JOIN ACTIVITIES a ON u.activityID = a.activityID
    WHERE f.muniID = m.muniID
    AND LOWER(a.activityName) LIKE '%music%'
);
```

```
SQL> SELECT COUNT(DISTINCT m.muniID) AS num_municipalities_without_music
2 FROM MUNICIPALITIES m
3 WHERE NOT EXISTS (
4     SELECT 1
5     FROM FACILITIES f
6     JOIN USES u ON f.facilityID = u.facilityID
7     JOIN ACTIVITIES a ON u.activityID = a.activityID
8     WHERE f.muniID = m.muniID
9     AND LOWER(a.activityName) LIKE '%music%'
10 );
```

NUM_MUNICIPALITIES_WITHOUT_MUSIC
2

1.5

```
SELECT DISTINCT p.provinceName  
  
FROM PROVINCES p  
  
JOIN MUNICIPALITIES m ON p.provinceID = m.provinceID  
  
WHERE m.avgPopulation >= 4000000;
```

```
SQL> SELECT DISTINCT p.provinceName  
2  FROM PROVINCES p  
3  JOIN MUNICIPALITIES m ON p.provinceID = m.provinceID  
4  WHERE m.avgPopulation >= 4000000;
```

```
PROVINCENAME
```

```
-----  
Gauteng
```

1.6

CREATE OR REPLACE PROCEDURE Province_Capacity_Utilization_procedure IS

BEGIN

FOR rec IN (

SELECT

p.provinceName AS province,

COUNT(DISTINCT f.facilityID) AS total_facilities,

SUM(f.capacity) AS total_capacity,

COUNT(u.activityID) AS total_activities,

ROUND((COUNT(u.activityID) / NULLIF(SUM(f.capacity), 0)) * 100, 2) AS
utilization_percentage

FROM PROVINCES p

JOIN MUNICIPALITIES m ON p.provinceID = m.provinceID

JOIN FACILITIES f ON m.muniID = f.muniID

LEFT JOIN USES u ON f.facilityID = u.facilityID

GROUP BY p.provinceName

ORDER BY utilization_percentage DESC

)

LOOP

DBMS_OUTPUT.PUT_LINE('Province: ' || rec.province);

DBMS_OUTPUT.PUT_LINE('Facilities: ' || rec.total_facilities ||

' | Capacity: ' || rec.total_capacity ||

' | Activities: ' || rec.total_activities ||

' | Utilization: ' || rec.utilization_percentage || '%');

DBMS_OUTPUT.PUT_LINE('-----');

END LOOP;

END;

/

```

SQL> CREATE OR REPLACE PROCEDURE Province_Capacity_Utilization_procedure IS
2 BEGIN
3   FOR rec IN (
4     SELECT
5       p.provinceName AS province,
6       COUNT(DISTINCT f.facilityID) AS total_facilities,
7       SUM(f.capacity) AS total_capacity,
8       COUNT(u.activityID) AS total_activities,
9       ROUND((COUNT(u.activityID) / NULLIF(SUM(f.capacity), 0)) * 100, 2) AS utilization_percentage
10      FROM PROVINCES p
11      JOIN MUNICIPALITIES m ON p.provinceID = m.provinceID
12      JOIN FACILITIES f ON m.muniID = f.muniID
13      LEFT JOIN USES u ON f.facilityID = u.facilityID
14      GROUP BY p.provinceName
15      ORDER BY utilization_percentage DESC
16   )
17   LOOP
18     DBMS_OUTPUT.PUT_LINE('Province: ' || rec.province);
19     DBMS_OUTPUT.PUT_LINE('Facilities: ' || rec.total_facilities ||
20       ' | Capacity: ' || rec.total_capacity ||
21       ' | Activities: ' || rec.total_activities ||
22       ' | Utilization: ' || rec.utilization_percentage || '%');
23     DBMS_OUTPUT.PUT_LINE('-----');
24   END LOOP;
25 END;
26 /

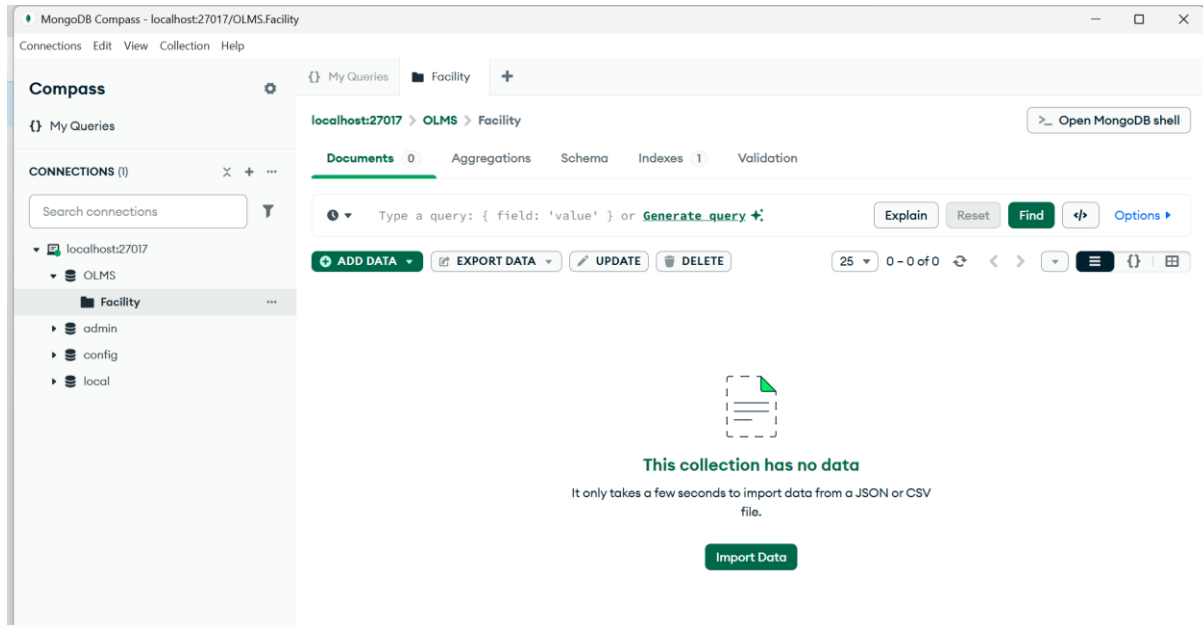
```

Procedure created.

Question 2

2.1

use OLMS



2.2

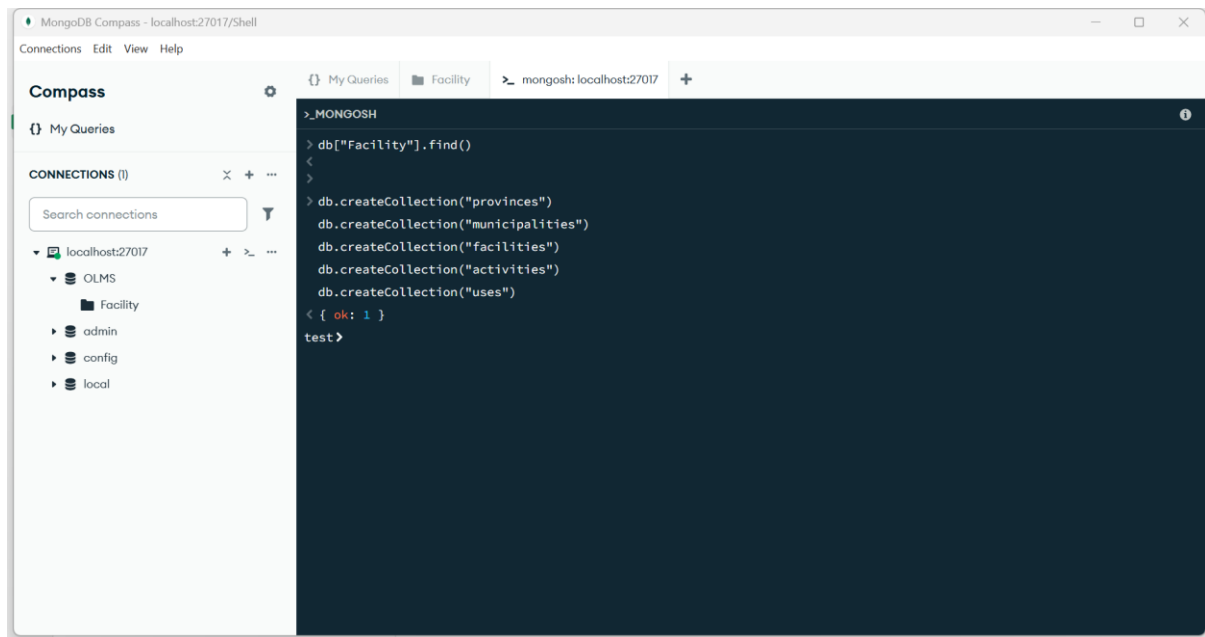
```
db.createCollection("provinces")
```

```
db.createCollection("municipalities")
```

```
db.createCollection("facilities")
```

```
db.createCollection("activities")
```

```
db.createCollection("uses")
```



2.3

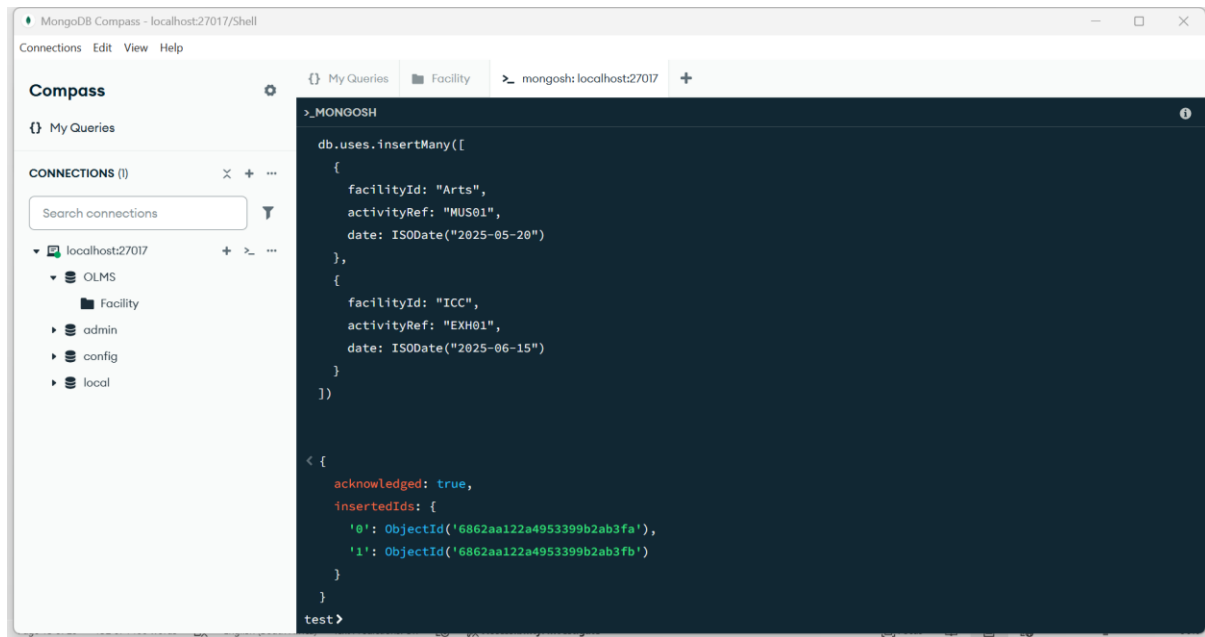
```
db.provinces.insertMany([
  {
    provinceCode: "WC",
    name: "Western Cape"
  },
  {
    provinceCode: "KZN",
    name: "KwaZulu-Natal"
  }
])
```

```
db.municipalities.insertMany([
  {
    munCode: "CPT",
    name: "City of Cape Town",
    avgPopulation: 4005016,
    provinceCode: "WC"
  },
  {
    munCode: "ETH",
    name: "eThekweni Metropolitan Municipality",
    avgPopulation: 3702231,
    provinceCode: "KZN"
  }
])
```

```
db.facilities.insertMany([
  {
    facilityId: "Arts",
    name: "Artscape Theatre Centre",
    address: "D.F. Malan St, Foreshore, Cape Town, 8001",
    capacity: 1500,
    munCode: "CPT",
    rooms: [
      { roomNo: 1, description: "Main Theatre" },
      { roomNo: 2, description: "Opera House" }
    ]
  },
  {
    facilityId: "ICC",
    name: "Durban ICC",
    address: "45 Bram Fischer Rd, Durban Central, Durban, 4001",
    capacity: 10000,
    munCode: "ETH",
    rooms: [
      { roomNo: 1, description: "Hall 1" },
      { roomNo: 2, description: "Hall 2" }
    ]
  }
])
```

```
db.activities.insertMany([
  {
    activityRef: "MUS01",
    name: "Music Concert"
  },
  {
    activityRef: "EXH01",
    name: "Art Exhibition"
  }
])
```

```
db.uses.insertMany([
  {
    facilityId: "Arts",
    activityRef: "MUS01",
    date: ISODate("2025-05-20")
  },
  {
    facilityId: "ICC",
    activityRef: "EXH01",
    date: ISODate("2025-06-15")
  }
])
```



2.4

```

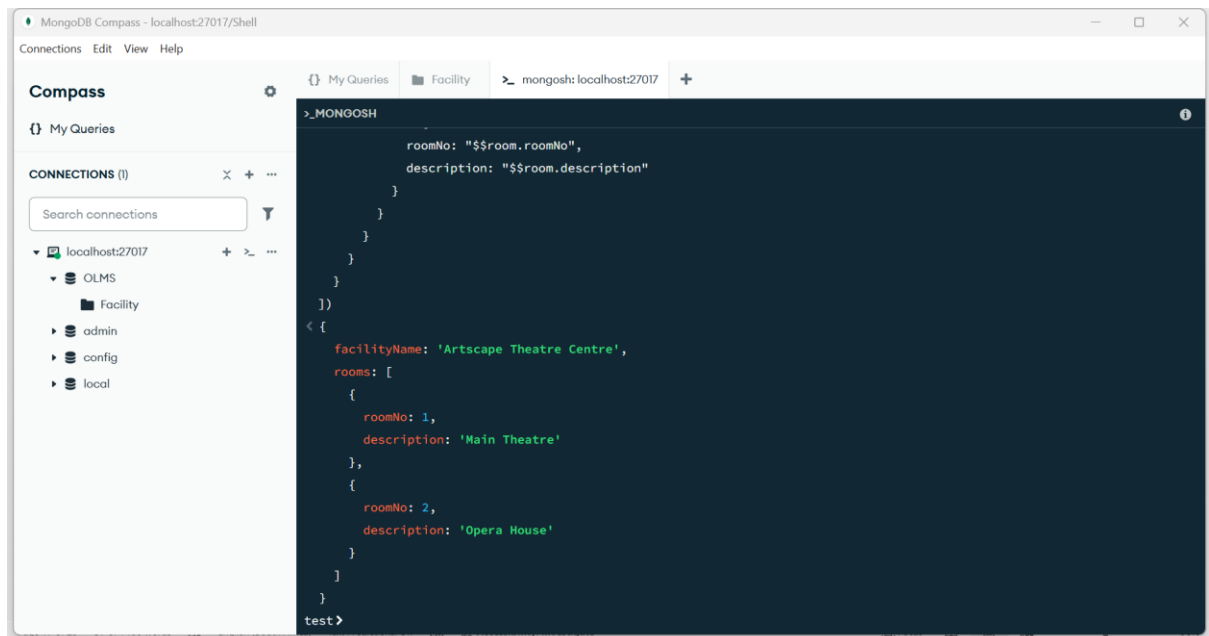
db.facilities.aggregate([
  {
    $match: { name: "Artscape Theatre Centre" }
  },
  {
    $project: {
      _id: 0,
      facilityName: "$name",
      rooms: {
        $map: {
          input: "$rooms",
          as: "room",

```

```

in: {
  roomNo: "$$room.roomNo",
  description: "$$room.description"
}
}
}
}
}
}
})

```



2.5

MongoDB Query

```

db.municipalities.find(
  { avgPopulation: { $gte: 3500000 } },

```

```
{
  _id: 0,
  munCode: 1,
  name: 1,
  avgPopulation: 1,
  provinceCode: 1
}
)
```

Sample Output

```
[
  {
    "munCode": "CPT",
    "name": "City of Cape Town",
    "avgPopulation": 4005016,
    "provinceCode": "WC"
  },
  {
    "munCode": "ETH",
    "name": "eThekweni Metropolitan Municipality",
    "avgPopulation": 3702231,
    "provinceCode": "KZN"
  }
]
```

```
< {
  munCode: 'CPT',
  name: 'City of Cape Town',
  avgPopulation: 4005016,
  provinceCode: 'WC'
}
{
  munCode: 'ETH',
  name: 'eThekweni Metropolitan Municipality',
  avgPopulation: 3702231,
  provinceCode: 'KZN'
}
```

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6862aa122a4953399b2ab3fa'),
    '1': ObjectId('6862aa122a4953399b2ab3fb')
  }
}
```

```
< {
  facilityName: 'Artscape Theatre Centre',
  rooms: [
    {
      roomNo: 1,
      description: 'Main Theatre'
    },
    {
      roomNo: 2,
      description: 'Opera House'
    }
  ]
}
```

2.6

```
db.uses.aggregate([
{
  $lookup: {
    from: "facilities",
    localField: "facilityId",
    foreignField: "facilityId",
    as: "facility_info"
  }
},
{
  $unwind: "$facility_info"
},
{
  $project: {
    _id: 0,
    date: 1,
    facilityName: "$facility_info.name",
    facilityCapacity: "$facility_info.capacity"
  }
}
])
```

```

< {
  date: 2025-05-20T00:00:00.000Z,
  facilityName: 'Artscape Theatre Centre',
  facilityCapacity: 1500
}
{
  date: 2025-06-15T00:00:00.000Z,
  facilityName: 'Durban ICC',
  facilityCapacity: 10000
}

```

2.7

```

db.provinces.insertMany([
  {
    provinceCode: "WC",
    provinceName: "Western Cape",
    municipalities: [
      {
        muniCode: "CPT",
        name: "City of Cape Town",
        avgPopulation: 4500000,
        facilities: [
          {
            facilityId: "ARTS",
            name: "Artscape Theatre Centre",
            capacity: 1500,

```

```
address: "D.F. Malan St, Foreshore, Cape Town",
rooms: [
  { roomNo: 1, description: "Main Theatre" },
  { roomNo: 2, description: "Opera House" }
],
uses: [
  {
    activity: {
      activityRef: "MUS01",
      name: "Music Concert"
    },
    date: ISODate("2025-05-20")
  }
]
}
]
}
],
{
  provinceCode: "KZN",
  provinceName: "KwaZulu-Natal",
  municipalities: [
    {
      muniCode: "ETH",
      name: "eThekweni Municipality",
      avgPopulation: 3700000,
      facilities: [
```



```
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6862aaec2a4953399b2ab3fc'),  
    '1': ObjectId('6862aaec2a4953399b2ab3fd')  
  }  
}
```

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SQL Tutorial – Full Database Course for Beginners. [Online] Available at: <https://www.youtube.com/watch?v=HXV3zeQKqGY>. [Accessed: 19 April 2024].

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Oracle® Database Express Edition Installation

Guide: <https://docs.oracle.com/en/database/oracle/oracle-database/18/xeinw/#Oracle%C2%AE-Database-Express-Edition>